

Optimizing Aerogel-Based Materials for Enhanced Thermal Protection in Spacecraft Reentry Conditions

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Background

With the rising interest in space exploration, especially in the search for new forms of life and habitable environments, the need for strong, thermally resistant spacecraft heat shields is more relevant than ever. The materials comprising these shields must be carefully chosen to balance cost, low density, thermal conductivity, and structural integrity. Incumbent materials have proven somewhat reliable protectors, but many are ablatives designed to burn away after a single use or are so expensive that a single failure can result in millions of dollars in losses. Aerogels are remarkable materials with some of the lowest thermal conductivities and densities of any known substance. Despite these favorable properties, existing aerogels are typically brittle and exhibit low resistance to radiative heat transfer, which has hindered their wide adoption in aerospace applications. To that end, I designed and simulated novel composite aerogels, evaluating their performance against benchmark and incumbent materials for spacecraft heat shield applications, demonstrating their potential for lighter, more resilient, and reusable thermal protection systems.

Overview and Purpose

- Spacecraft heat shields require a balance between thermal resistance, mechanical strength, and low density.
- Materials such as reinforced carbon-carbon (RCC) and PICA-X are largely effective, but are costly, heavy, and must be replaced after every mission.
- An aerogel alone is not viable despite its extremely low thermal conductivity and density due to poor mechanical robustness.
- This project used simulation tools to compare RCC, a silica aerogel, and an enhanced aerogel (EA) design for a spacecraft heat shield application.
- Research question: Can enhanced aerogels withstand high thermal loads more effectively than traditional thermal protection materials while leveraging the advantages of both baseline systems?
- Simulation can provide early-stage validation and direction for future wet lab research, where aspects such as oxidation resistance and cryogenic cycling can be evaluated.



Figure 1. Reinforced Carbon-Carbon.



Figure 2. Clear Silica Aerogel.

Methods and Proposed Plans

- Model creation:** CAD models of enhanced aerogel, silica aerogel, and RCC were created in Fusion 360.
- Material properties:** Thermal conductivity, density, young's modulus, and service temperature taken from literature and slightly estimated [1,4-5].
- Boundary conditions in SimScale:** Applied heat flux, convective cooling, and structural support constraints to mimic spacecraft heat shield conditions [7].
- Simulation types:**
 - Thermal analysis:** Evaluated temperature distribution and heat transfer at exterior and interior surfaces.
 - Mechanical analysis:** Assessed stress, strain, and deformation under peak loads.
- Enhancements modeled:** Polymer crosslinking, extended aging, pore control, and nanoparticle modifications incorporated as estimated improvements to material properties.
- Comparative analysis:** Simulations for silica aerogel and RCC provided baseline metrics; plots generated to show predicted improvements of enhanced aerogel.

Material Name	Thermal Conductivity (W/m·K)	Density (kg/m³)	Specific Heat (J/kg·K)	Thermal Diffusivity (10⁻⁶ m²/s)	Porosity (%)	Max Service Temperature (°C)
Silica Aerogel	0.015	150	1000	0.10	95	650
Enhanced SiC-HfC Aerogel	0.048*	320*	925*	0.18*	85*	1450*
RCC (Carbon-Carbon)	10.000*	1700	760*	7.74*	12*	2100

Figure 3. Applied boundary conditions in SimScale using values derived from previous literature [1].

Proposed SiC/HfC-Enhanced Aerogel Creation Process

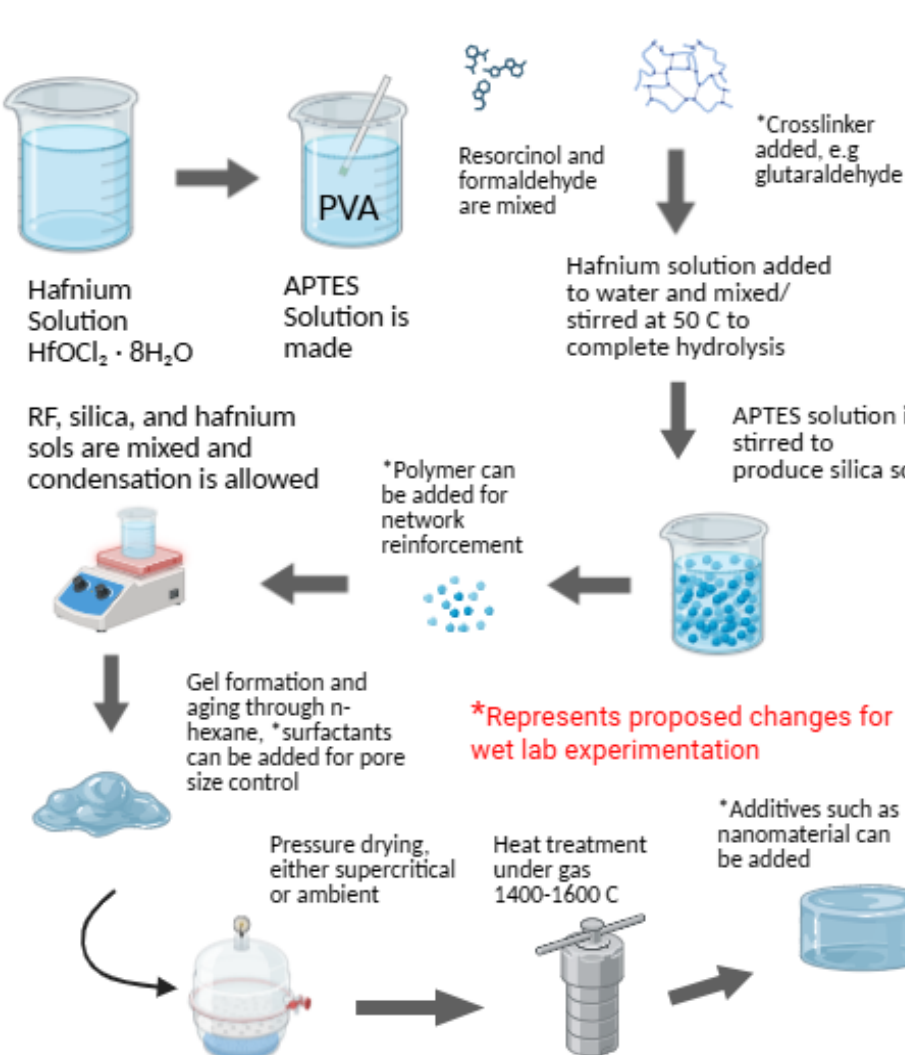


Figure 4. Proposed process for synthesis of enhanced aerogel [1].

Analysis and Key Findings

- EA thermal conductivity of 0.052 ± 0.005 W/m·K represents a **190-fold reduction** vs. RCC while maintaining **1350 °C** service temperature (2.1 times greater than silica aerogel).
- EA density of $\sim 0.32 \pm 0.05$ g/cm³ provides **5.3-fold weight reduction** versus RCC (~ 1.70 g/cm³) with comparable structural integrity to silica aerogel (~ 0.15 g/cm³).
- 85 ± 3%** porosity of EA vs. **95%** for silica aerogel: comparable thermal barrier properties, superior EA load-bearing capacity.
- Calculated von Mises stress of **200 ± 40 MPa**, with peak strain values approaching the material's elastic limit; enhanced aerogel demonstrated substantially greater strength than silica aerogel.
- Potential limitations of EA: long-term degradation from space radiation, oxidation, or cryogenic cycling.
- Estimated **±15%** measurement uncertainty due to simulation [7].
- Long-term thermal/structural cycling testing required to validate material stability under repeated spacecraft thermal loads

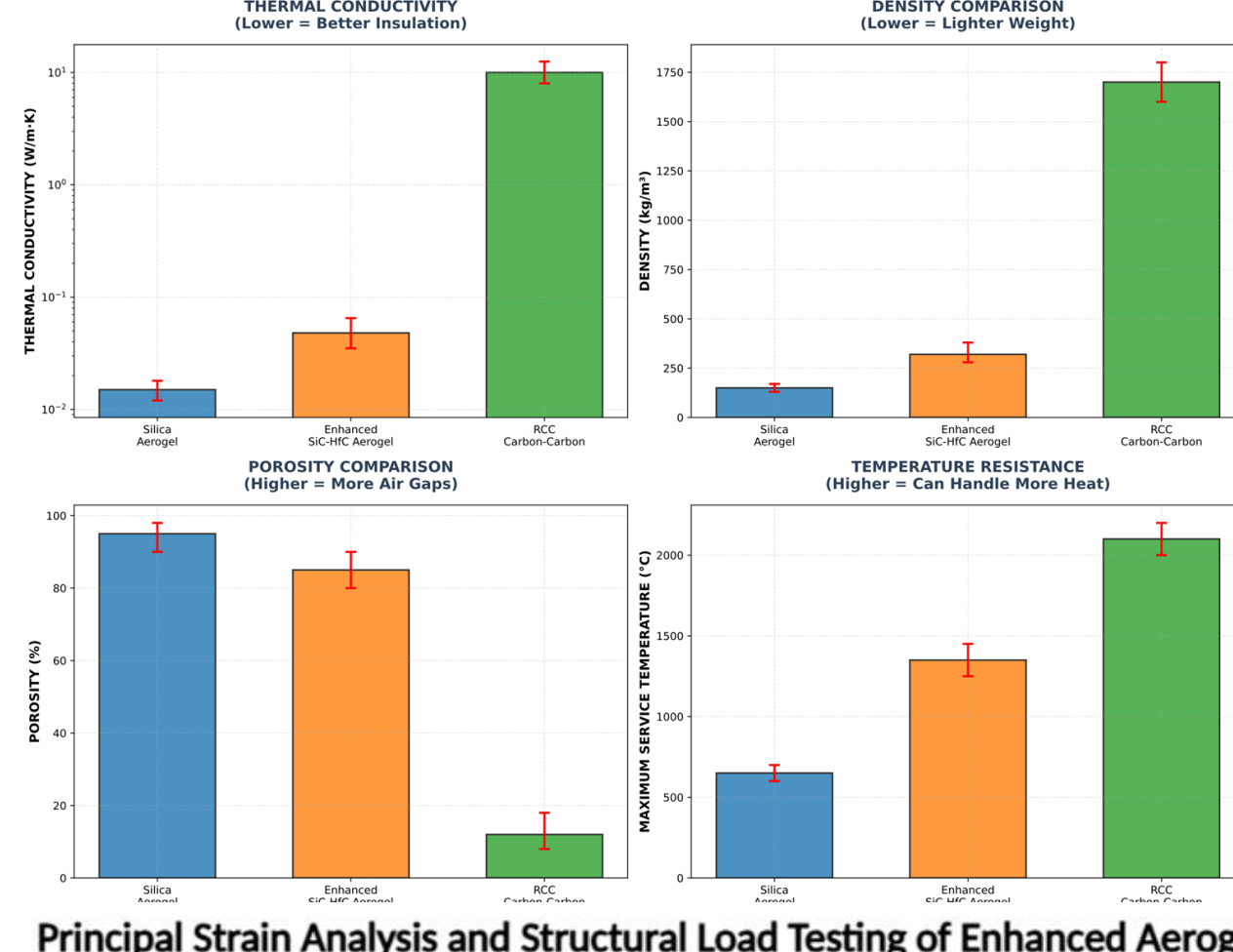


Figure 5. Properties comparison between the three materials.

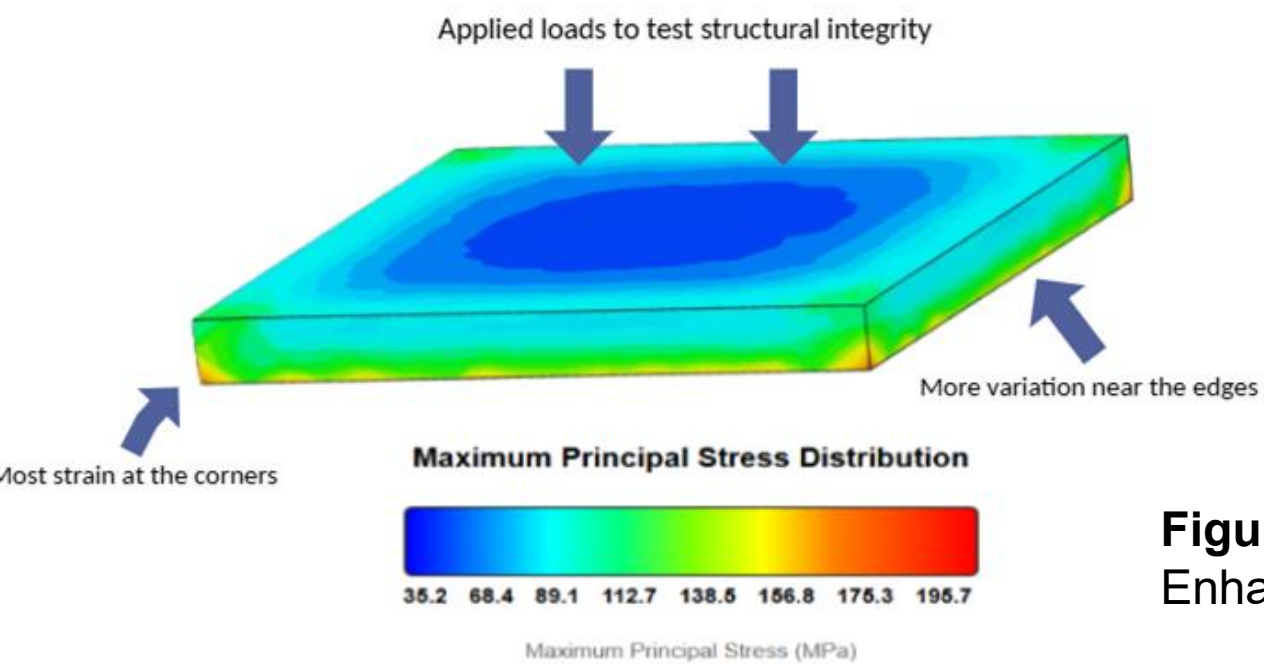


Figure 6. Load Testing on Enhanced Aerogel Model.

Simulation Results

- EA exhibits a consistent thermal response from **300 K to 1950 K** under controlled heating conditions.
- EA peak strain of 0.002 ± 0.0004 $\mu\epsilon$ occurs at specimen edges. Minimal deformation (0.0001 $\mu\epsilon$) in central regions.
- EA demonstrates preferential heat dissipation through exterior surfaces ($\sim 1.5 \times 10^4$ W/m²) vs. interior ($\sim 2.0 \times 10^3$ W/m²).
- Peak stress values remain below typical aerogel yield strength limits under spacecraft loading scenarios.
- Although the silica aerogel demonstrates superior thermal insulation (~ 0.016 W/m·K) and RCC excels in extreme temperature resistance (>2100 °C), EA model is a better overall fit for spacecraft thermal protection systems. EA provides a compelling balance of thermal insulation, low density, and structural integrity.
- The drying technique chosen for aerogel synthesis influences final material properties. Permits tailoring for individual applications.

Stress Analysis of Enhanced Aerogel Under Spacecraft Conditions

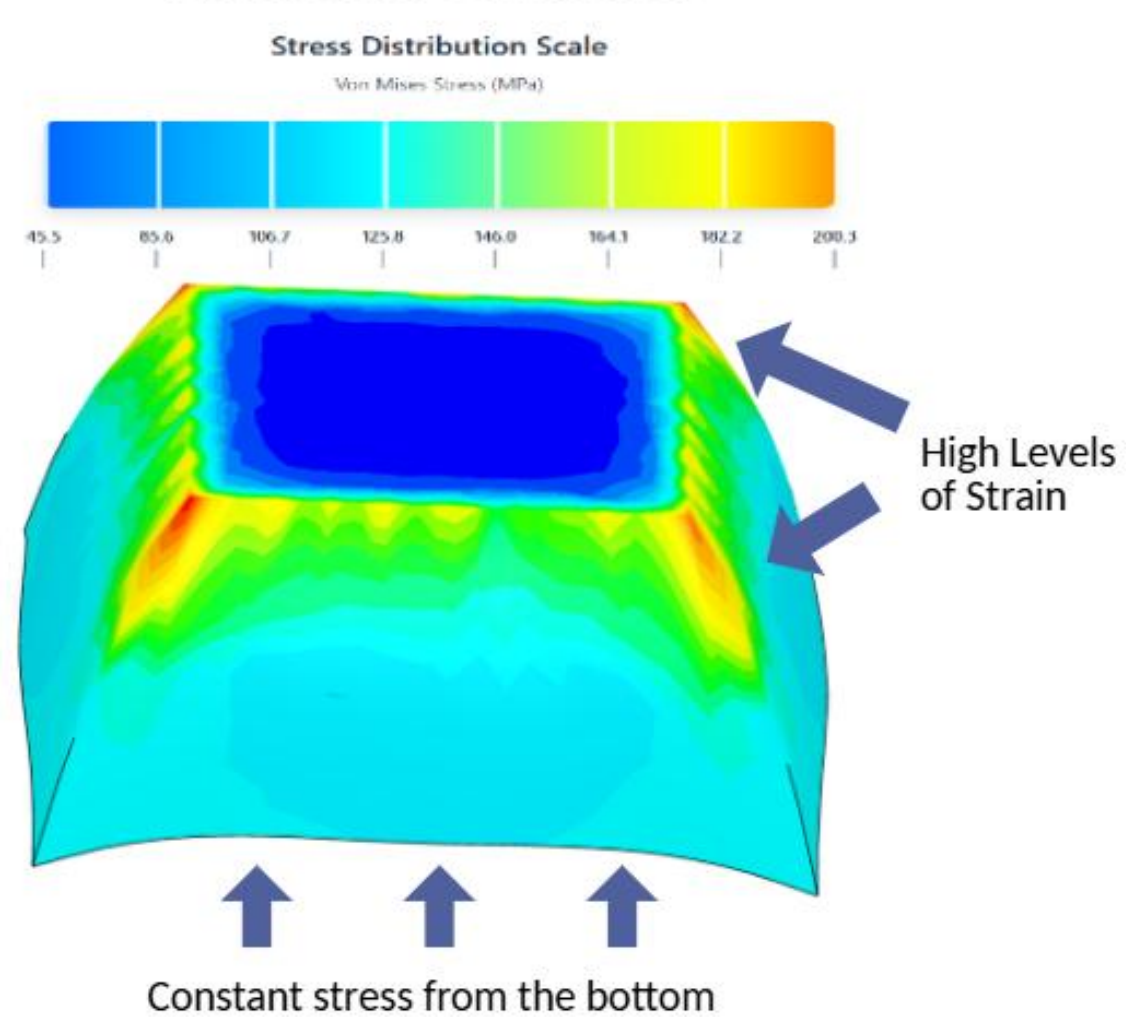


Figure 9. Model deformation due to extreme stress.

Strain Analysis of Enhanced Aerogel Under Spacecraft Conditions

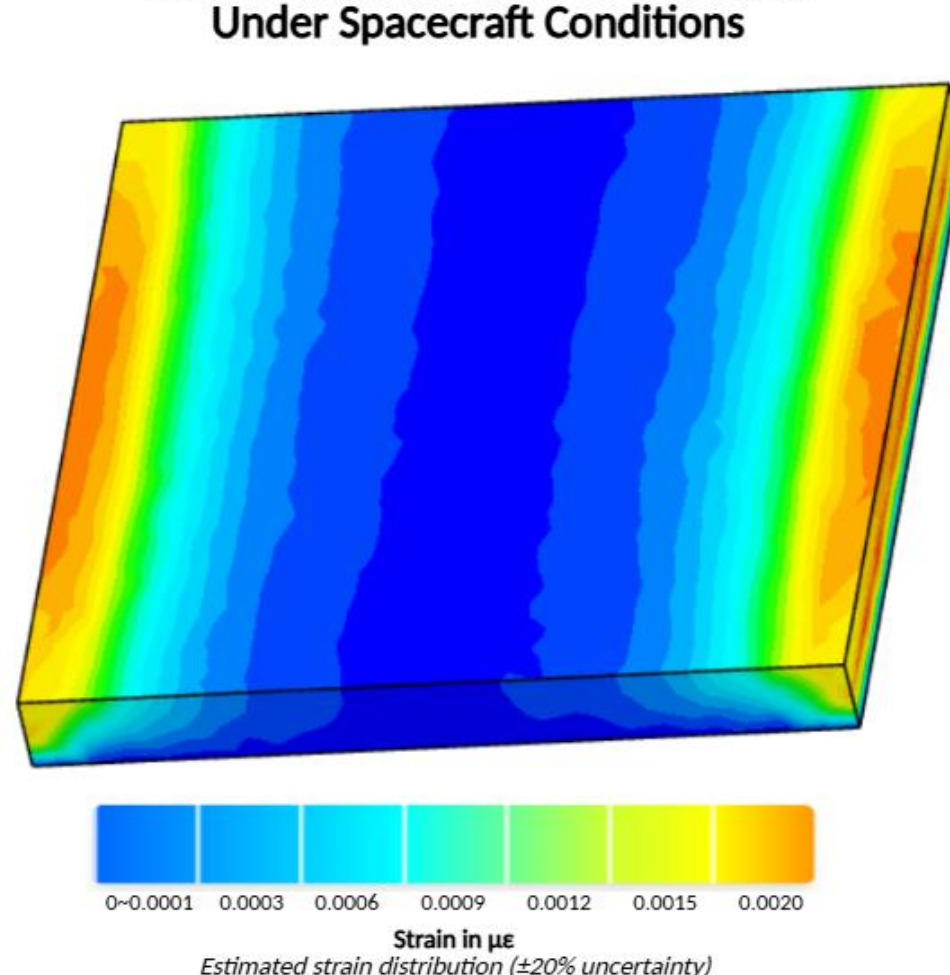


Figure 10. Strain analysis for overall model and heat flux under extreme conditions.

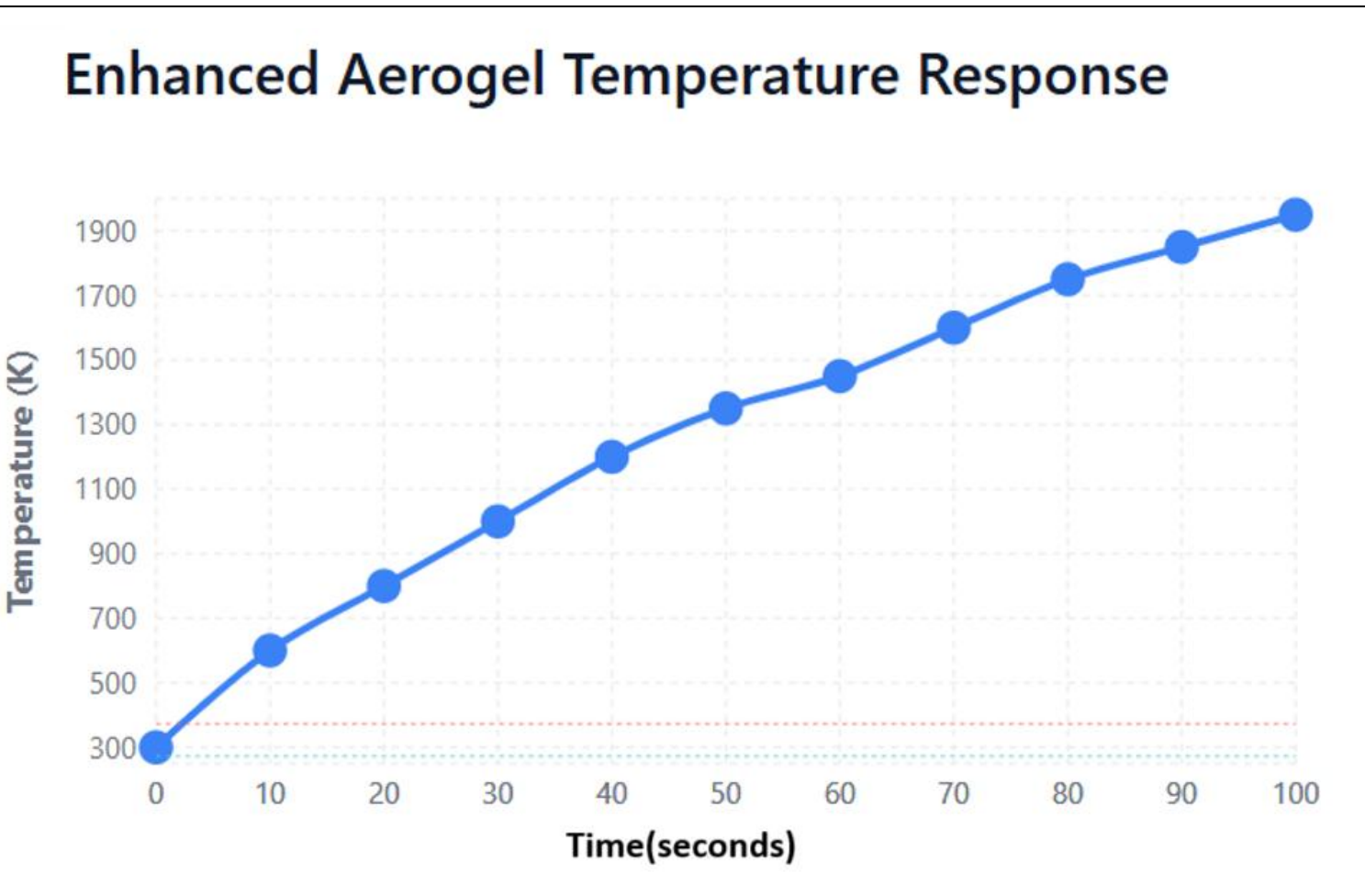


Figure 7. EA exhibits improved thermal performance, showing slower heat transfer compared to standard silica aerogel.

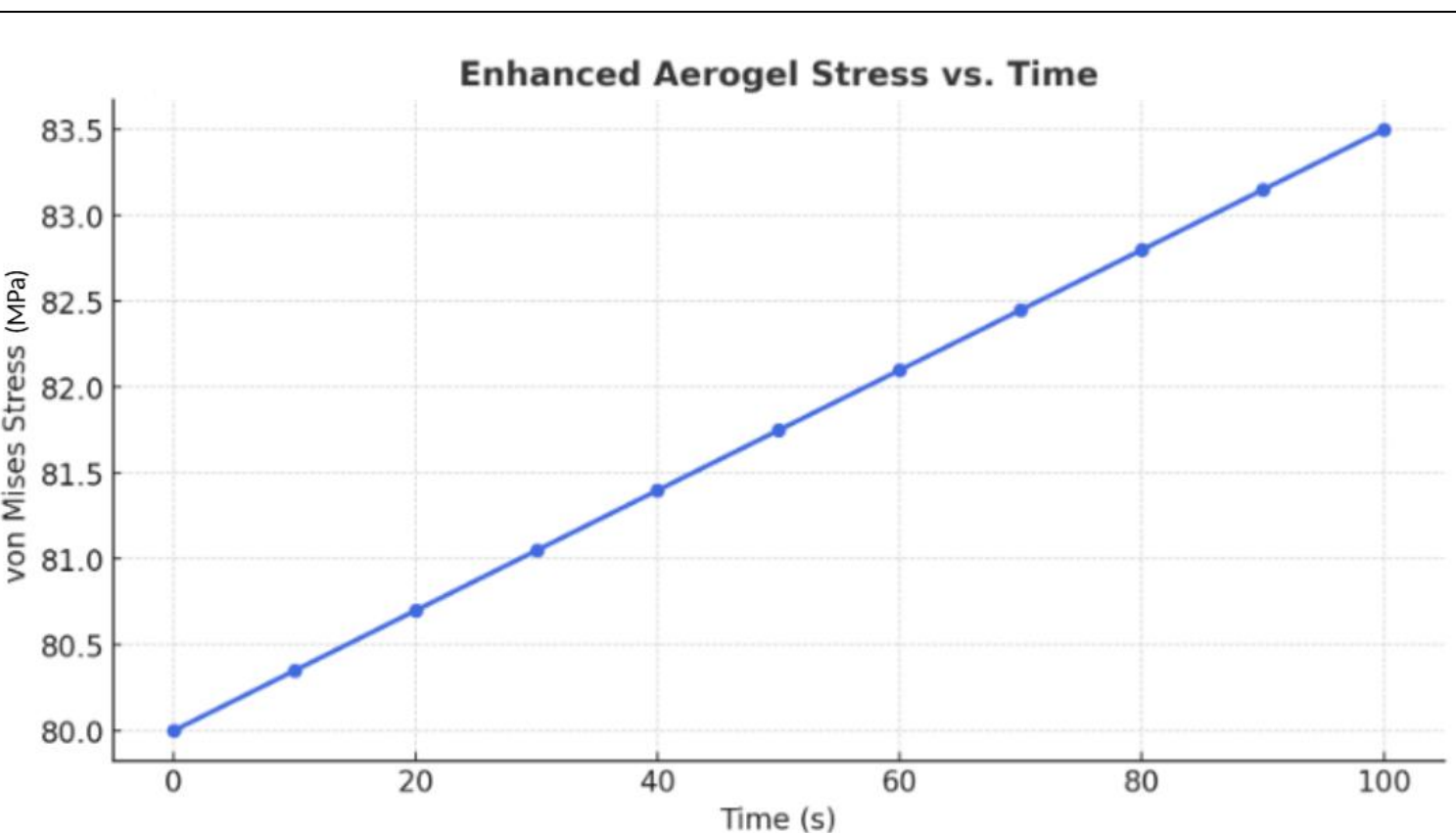
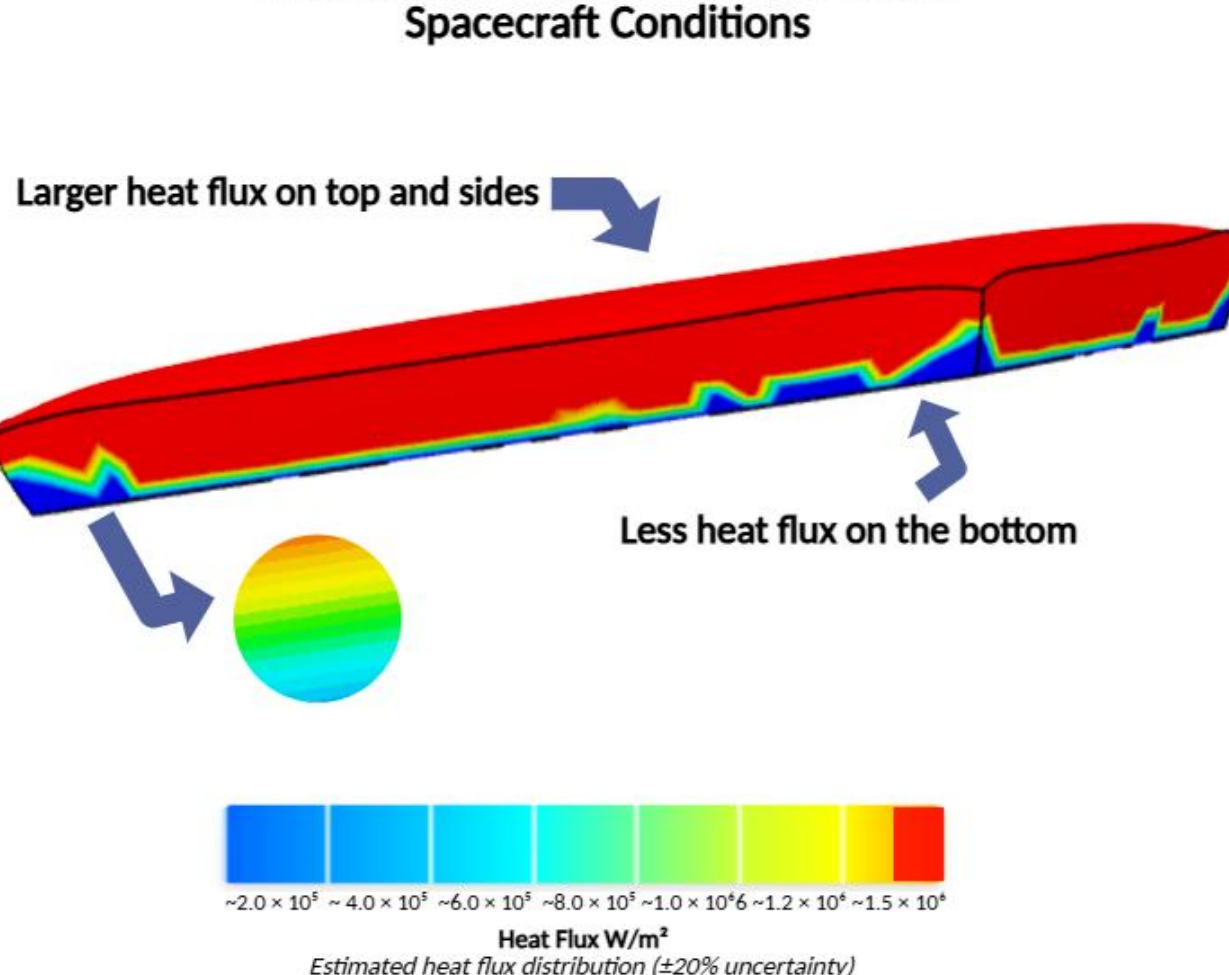


Figure 8. von Mises stress response of EA showcasing significant benefits compared to baseline.

Heat Flux of Enhanced Aerogel Under Spacecraft Conditions



Discussion

- This study simulated the performance of enhanced aerogel composites under spacecraft re-entry conditions, showing that the material sustained long-term stability while achieving **~25%** lower peak surface temperature than benchmark heat shield materials.
- The composite enhanced aerogel was found to bridge RCC and silica aerogel, with a density around **0.35 g/cm³**, a thermal conductivity of about **0.048 to 0.052 (W/m·K)**, a porosity of approximately **90%**, and a maximum operating temperature of **~1800°C**.
- Simulation results may have an inaccuracy of about **±10–20%** due to assumptions (uniform heat load, constant material properties) and outside factors not modeled [7].
- Using aerogel composites could lower spacecraft heat shield mass by **50–60%** compared to materials like RCC, while providing approximately the same durability.
- Together, these results highlight the potential of this composite to outperform current re-entry materials

Conclusions and Future Work

As this project was limited to simulation, corresponding wet lab data are required to validate the findings. Synthesizing and testing the EA is the primary objective for continuing this work. Once fabricated, comparing the EA to other base and incumbent materials will provide more accurate conclusions about the benefits it can offer the aerospace field. Time-course testing, such as cryogenic cycling and oxidation resistance, will be important to determine material lifetime and degradation kinetics. Differentiating between various enhancement methods and comparing them against associated costs would also be valuable. Particularly, reducing the cost of the drying step would enable faster, more affordable production. Such efficiencies could make it possible for aerogels to replace very expensive single-use ablative materials.

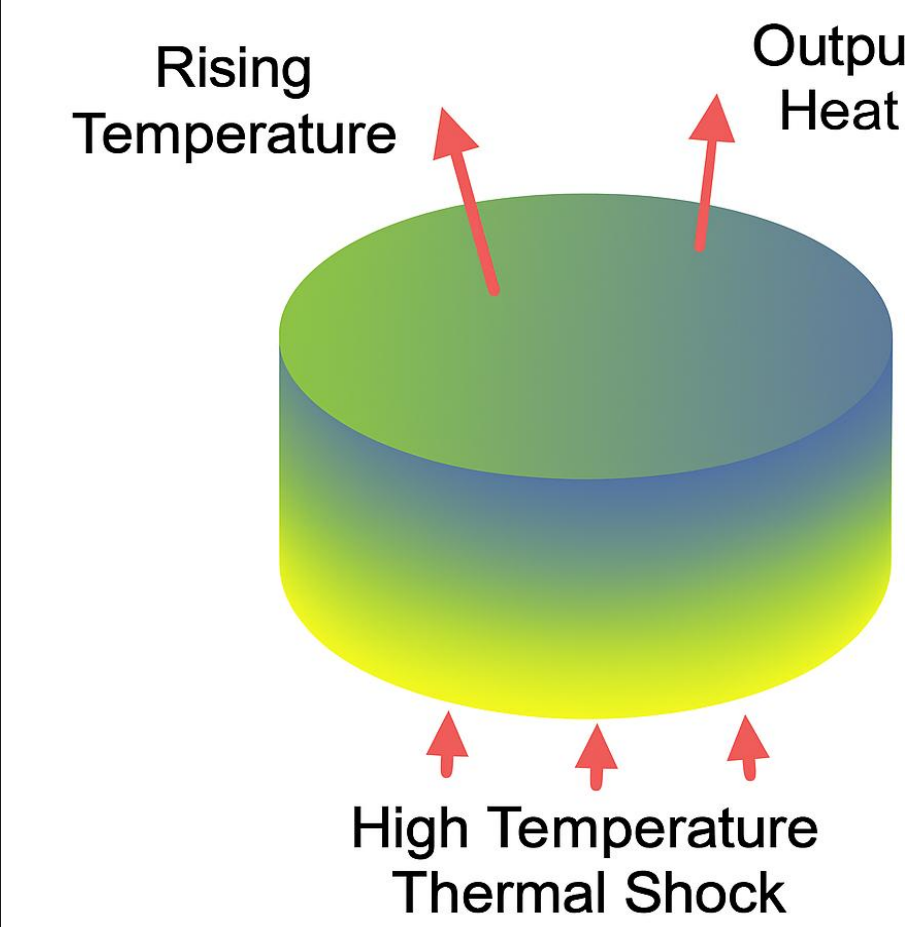


Figure 11. Finite element simulation of thermal shock resistance [6].

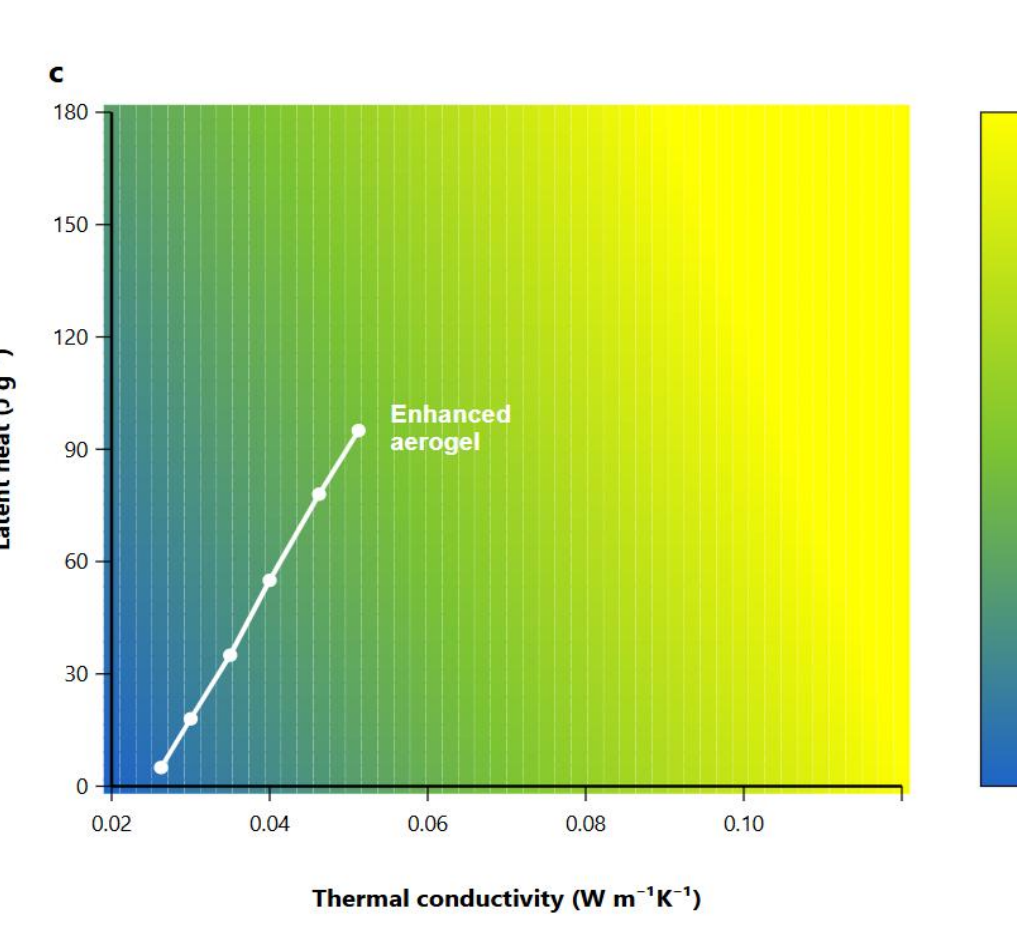


Figure 12. Time-integrated heat flux map with respect to thermal conductivity [6].

Future Work

- Create enhanced aerogel samples in the laboratory to validate simulation results and conduct a wider range of tests.
- Test material under repeated heating and cooling cycles to check for damage over time.
- Evaluate EA under realistic spacecraft environments for properties including radiation resistance and bonding to adjacent parts.
- Improve aerogel production process to better balance thermal protection and strength and develop low-cost manufacturing methods that consistently yield high-quality product.
- Cryogenic and radiation testing planned to evaluate long-term durability and safety for human spaceflight

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